



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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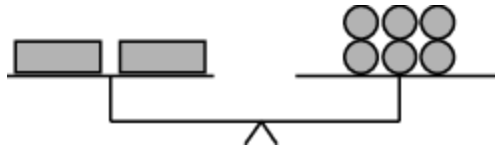
**Gauss Contest Grade 7
Problems**

1. The value of $(2 + 4 + 6) - (1 + 3 + 5)$ is
 (A) 0 (B) 3 (C) -3 (D) 21 (E) 111
-

2. $0.8 - 0.07$ equals
 (A) 0.1 (B) 0.71 (C) 0.793 (D) 0.01 (E) 0.73
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3. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8}$ is equal to
 (A) 1 (B) $\frac{1}{64}$ (C) $\frac{3}{14}$ (D) $\frac{7}{8}$ (E) $\frac{3}{8}$
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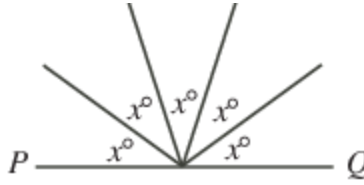
4. The equal-arm scale shown is balanced. One  has the same mass as



- (A)  (B)  (C) 
 (D)  (E) 
-

5. If there is no tax, which of the following costs more than \$18 to purchase?
 (A) Five \$1 items and five \$2 items
 (B) Nine \$1 items and four \$2 items
 (C) Nine \$1 items and five \$2 items
 (D) Two \$1 items and six \$2 items
 (E) Sixteen \$1 items and no \$2 items
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6. If PQ is a straight line, then the value of x is



- (A) 36 (B) 72 (C) 18 (D) 20 (E) 45
-

7. $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$ equals

- (A) $3\frac{1}{3}$ (B) $7 + \frac{1}{3}$ (C) $\frac{3}{7}$ (D) $7 + 3$ (E) $7 \times \frac{1}{3}$
-

8. A circle has a radius of 4 cm. A line segment joins two points on the circle. What is the greatest possible length of the line segment?

- (A) 10 cm (B) 8 cm (C) 4 cm (D) 12 cm (E) 6 cm
-

9. The word **BANK** is painted exactly as shown on the outside of a clear glass window. Looking out through the window from the inside of the building, the word appears as

- (A) **BANK** (B) **KNAB** (C) **BAK** (D) **KNAB** (E) **KNAB**
-

10. Which one of the following is equal to 17?

- (A) $3 - 4 \times 5 + 6$ (B) $3 \times 4 + 5 \div 6$ (C) $3 + 4 \times 5 - 6$
 (D) $3 \div 4 + 5 - 6$ (E) $3 \times 4 \div 5 + 6$
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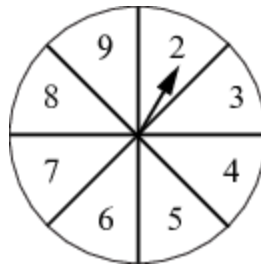
11. The sum of the prime factors of 42 is

- (A) 23 (B) 43 (C) 12 (D) 17 (E) 13
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12. A bamboo plant grows at a rate of 105 cm per day. On May 1st at noon it was 2 m tall. Approximately how tall, in metres, was it on May 8th at noon?
- (A) 10.4 (B) 8.3 (C) 3.05 (D) 7.35 (E) 9.35
-

13. If the mean (average) of five consecutive integers is 21, the smallest of the five integers is
- (A) 17 (B) 21 (C) 1 (D) 18 (E) 19
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14. A circular spinner is divided into 8 equal sections, as shown. An arrow is attached to the centre of the spinner. The arrow is spun once. What is the probability that the arrow stops in a section containing a prime number that is odd?



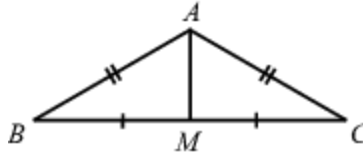
- (A) $\frac{1}{8}$ (B) $\frac{2}{8}$ (C) $\frac{3}{8}$ (D) $\frac{4}{8}$ (E) $\frac{7}{8}$
-

15. Suppose n is a number such that the mean (average) of the list of numbers 2, 9, 4, n , $2n$ is equal to 6. What is the value of n ?
- (A) 9 (B) 12 (C) 10 (D) 5 (E) 6
-

16. A standard fair coin is tossed three times. What is the probability that the three outcomes are all the same?
- (A) $\frac{1}{2}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{5}{16}$ (E) $\frac{1}{8}$
-

17. The ratio of junior kindergarteners to senior kindergarteners at Gauss Public School is 8 : 5. If there are 128 junior kindergarteners at the school, then how many kindergarteners are there at the school?
- (A) 218 (B) 253 (C) 208 (D) 133 (E) 198
-

18. In the diagram, $\triangle ABC$ is isosceles. M is on BC so that $BM = MC$. If the perimeter of $\triangle ABC$ is 64 and the perimeter of $\triangle ABM$ is 40, what is the length of AM ?



- (A) 10 (B) 8 (C) 16 (D) 12 (E) 24

19. Which of the following is closest to one million (10^6) seconds?
 (A) 1 day (B) 10 days (C) 100 days (D) 1 year (E) 10 years

20. The letter P is written in a 2×2 grid of squares as shown:  A combination of rotations about the centre of the grid and reflections in the two lines through the centre achieves the result: 

When the same combination of rotations and reflect is applied to , the result is

- (A)  (B)  (C)  (D)  (E) 

21. Kathy owns more cats than Alice and more dogs than Bruce. Alice owns more dogs than Kathy and fewer cats than Bruce. Which of the statements *it must* be true?
 (A) Bruce owns the fewest cats.
 (B) Bruce owns the most cats.
 (C) Kathy owns the most cats.
 (D) Alice owns the most dogs.
 (E) Kathy owns the fewest dogs.

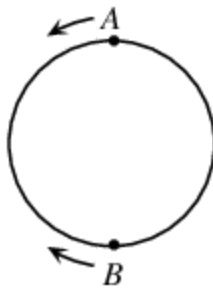
22. In a bin at the Gauss Grocery, the ratio of the number of apples to the number of oranges is $1 : 4$, and the ratio of the number of oranges to the number of lemons is $5 : 2$. What is the ratio of the number of apples to the number of lemons?
- (A) $1 : 2$ (B) $4 : 5$ (C) $5 : 8$ (D) $20 : 8$ (E) $2 : 1$
-

23. In the addition of two 2-digit numbers, each blank space, including those in the answer, is to be filled with one of the digits 0, 1, 2, 3, 4, 5, 6, each used exactly once. The units digit of the sum is

$$\begin{array}{r}
 \square \square \\
 + \square \square \\
 \hline
 \square \square ?
 \end{array}$$

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6
-

24. On a circular track, Alphonse is at point A and Beryl is diametrically opposite at point B . Alphonse runs counterclockwise and Beryl runs clockwise. They run at constant, but different, speeds. After running for a while they notice that when they pass each other it is always at the same three places on the track. What is the ratio of their speeds?



- (A) $3 : 2$ (B) $3 : 1$ (C) $4 : 1$ (D) $2 : 1$ (E) $5 : 2$
-

25. A school trip offered its participants three activities: hiking, canoeing and swimming. Attendance records show that of all participants
- 10 students participated in all three activities,
 - 50% participated in at least hiking and canoeing,
 - 60% participated in at least hiking and swimming,
 - $k\%$ participated in at least canoeing and swimming, and
 - no students participated in fewer than two activities.

If k is a positive integer, what is the sum of all possible values of k ?

- (A) 191 (B) 185 (C) 261 (D) 95 (E) 175
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